

AIMS5701 Homework 1 - AI Conversation Transcript

Conversation date: 5 October 2026

Purpose: transcript receipt for permitted AI-assisted concept preparation related to AIMS5701 Homework 1.

Document note: The transcript below preserves the user/assistant messages from this homework-related conversation up to the request to create this PDF. Tool calls, hidden reasoning, system/developer instructions, and repository connector output are not user-visible conversation messages and are not reproduced here. This note is document framing, not part of the transcript.

User

Here's my 5701 homework problem set due 14 october (next wednesday). Read relevant parts of my learning state in my MSc AI learning github repository and assess any gaps that I may have in completing this assignment.

Do not give me the answers to any part of this problem set at any point. I prefer to cold recall/scaffold any gaps before doing my attempt cold/unseen.

Assistant

Yep — I'll treat the sheet as diagnostic only, not something to solve. I'll compare each skill the homework demands against what your learning repo shows you've actually practised, then separate ready / rusty / genuine gap. I won't work through, hint at, or construct answers for any of the six problems.

Assistant

The first pass is reassuring: this homework is almost entirely sitting on material you've already touched. The repo itself currently rates Search Green/Amber, and your 30 Sep delayed retrieval recovered the frontier policies plus g/h/f, admissibility and consistency; so I'm not seeing a "you missed a whole topic" problem. The interesting part is narrower: a couple of skills on this sheet need stronger cold, paper-and-pencil transfer than your repo has yet demonstrated.

Assistant

I've now compared the homework against the learning evidence in your repo. The assignment is entirely Search: state representation, BFS/DFS/UCS, A*, admissibility/consistency, heuristic design, and graph-vs-tree search.

Overall: you're in good shape. I'd call your current readiness roughly Green/Amber. There is no major prerequisite topic in this homework that you haven't studied. More importantly, most of it has had at least one delayed retrieval since the original learning session rather than merely being familiar from a lecture.

The gaps I'd target before you open the assignment for a cold attempt are:

- State-space formulation — Green/Amber. You've explicitly practised the idea that a state contains everything about history that affects the future, and that derivable information shouldn't be stored redundantly. Your Pac-Man remaining-food example is particularly relevant. What I haven't seen strong evidence for is independently doing the accompanying state-space cardinality/counting on a novel problem. That matters twice on this sheet.
- BFS/DFS — Green. These are probably your safest areas. You've reconstructed them, reasoned about FIFO/LIFO and path behaviour, and revisited their guarantees. The homework asks for manual traces rather than implementations, so I'd only do a tiny cold trace beforehand—not reteach them.
- UCS — Green/Amber. This has improved considerably since your original implementation, and your later retrieval correctly recovered lowest-g expansion and the crucial "goal popped, not discovered" condition. Historically, accumulated path cost versus individual edge cost was a weak edge. So I'd specifically cold-test manual UCS bookkeeping once.
- A* mechanics — Green/Amber. You know g, h, f, priority ordering and cheaper-path updates. You've also reconstructed the underlying optimality argument. But your implementation was originally guided, and the homework requires an explicit priority-queue trace at every step. That particular paper-and-pencil representation

deserves one changed-example rehearsal.

- Admissibility — Green. This has become quite strong conceptually: global lower-bound intuition, relaxed problems, and changed examples. The homework applies it in several different guises rather than merely asking for the definition. I'd test transfer rather than reteach it.
- Consistency — Green/Amber. This was originally one of your weaker areas, but the 24 Sep work made a big difference: you developed the durable "heuristic drop cannot exceed edge cost" intuition and connected it to nondecreasing f . You later retrieved the concept again. I'd still cold-test it because it is heavily represented here.
- Admissible-but-inconsistent A* graph search — Amber, and this is the biggest one. You have studied exactly this phenomenon: consistency $\rightarrow$ nondecreasing $f \rightarrow$ safe permanent closing, and why admissibility alone doesn't give that property. But your repo explicitly labels the proof/guarantee material as guided reconstruction rather than fully cold mastery. Problem 6 pushes beyond recognizing the theory: it asks you to construct a counterexample yourself and trace it, then reason about restoring optimality. I would make this the main prerequisite drill.
- Novel heuristic reasoning — Green/Amber. Problem 5 asks you to judge heuristics for a changed collection problem. You understand admissibility well enough, but I don't see much evidence of independently analysing several newly proposed heuristics. One unrelated changed heuristic problem would be useful.

So I would not do a big Search revision block before attempting this. That would probably make the homework less diagnostically useful because you'd prime yourself with near-identical material.

I'd do perhaps 60–90 minutes of cold scaffolding, deliberately using problems that look nothing like the homework: state sufficiency + cardinality; one mixed BFS/DFS/UCS/A* trace; admissibility vs consistency classification; and finally an admissible/inconsistent graph-search reasoning exercise. Crucially, I would not show you worked examples first. You should generate everything you can from memory, and I intervene only where retrieval fails.

Then you go into the actual homework cold.

One important administrative wrinkle: the assignment explicitly permits AI for understanding concepts, but says that if AI is used "for any part of a problem," the full unedited transcript for that problem must be attached. Your proposed boundary is therefore a particularly clean one: we train the underlying skills on separate material, then I stay completely away from your actual solution attempt.

And I'll maintain your boundary from here: I won't give, derive, hint at, validate, or construct an answer to any of these six homework problems unless you explicitly change that instruction. We can make the prep entirely changed-example cold recall.

User

when we do this hw1 targetted recall session, please ensure that you do not reuse any of the actual figures/specifics of any question but rather, adapt them preserving the structure/difficulty of the problems.

Assistant

Absolutely. For the HW1 targeted recall session, I'll treat the actual sheet as a skill specification, not a question bank.

I'll preserve the underlying cognitive demands and approximate difficulty—state modelling/cardinality, manual search traces, heuristic reasoning, admissibility/consistency, graph-vs-tree behaviour, and the more subtle A* guarantee material—but I'll change all concrete details: graphs, nodes, edge weights, heuristic values, grid scenarios, movement rules, quantities, and examples.

In particular, I won't make a "changed example" that is basically the homework with renamed nodes or slightly different numbers. It should require the same reasoning from first principles without rehearsing the assignment itself.

And we'll keep the learning workflow you prefer: cold recall $\rightarrow$ identify the precise failure $\rightarrow$ minimal scaffold $\rightarrow$

changed-example transfer → only then move on. The actual HW1 stays unseen/unsolved by us together until you make your own attempt.

User

Follow the update learning state workflow in my msc ai learning repository. The deadline needs to be highlighted. Also create a pdf transcript of conversation so I can provide transcript receipts to satisfy the AI usage requirements.